INFORMATIQUE

CHAPTER 1 - INTRODUCTION

Objective and plan

- **Objective :**
  - Prerequisites and upgrading
- **Plan :**
  - Introduction
  - Material
  - Operating system
  - Computer languages
  - Applications
  - Data representation
  - Data size measurement units

Introduction to Computer Science

- **Definition: Techniques for the automatic processing of information using computers**

- **Information processing**
  - Calculate
  - Transform
  - Dematerialize
  - Store
  - Transfer

Introduction to Computer Science

Elements of a Computer System

Hardware
(PC, Mac, Tablet, Workstation, etc...)

Operating system
(Windows , Mac OS, Android , Linux, etc...)

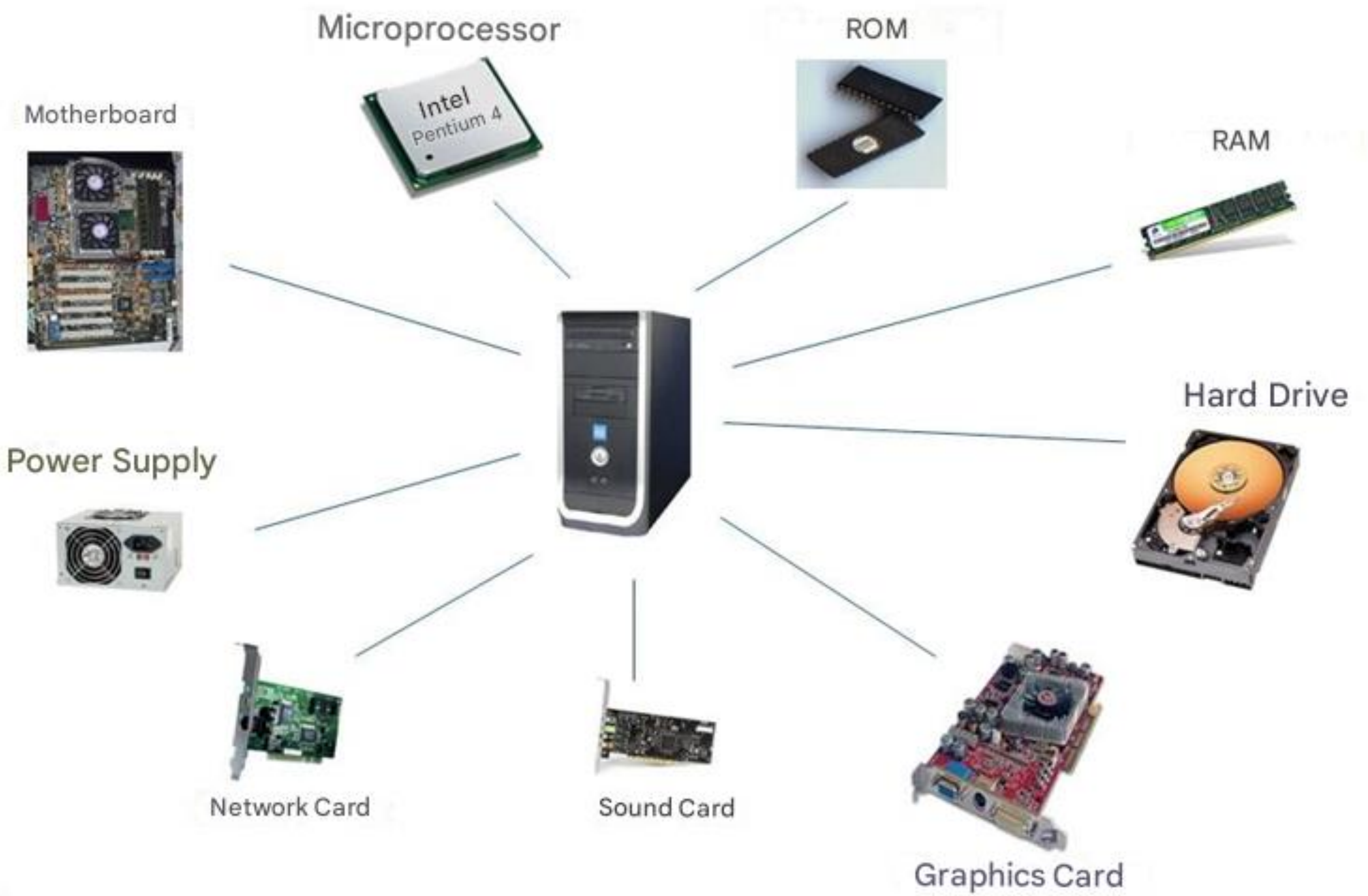
Languages
(Java, C/C++, Fortran, Python, etc.)

Applications
(Word, Excel, Games, etc…)

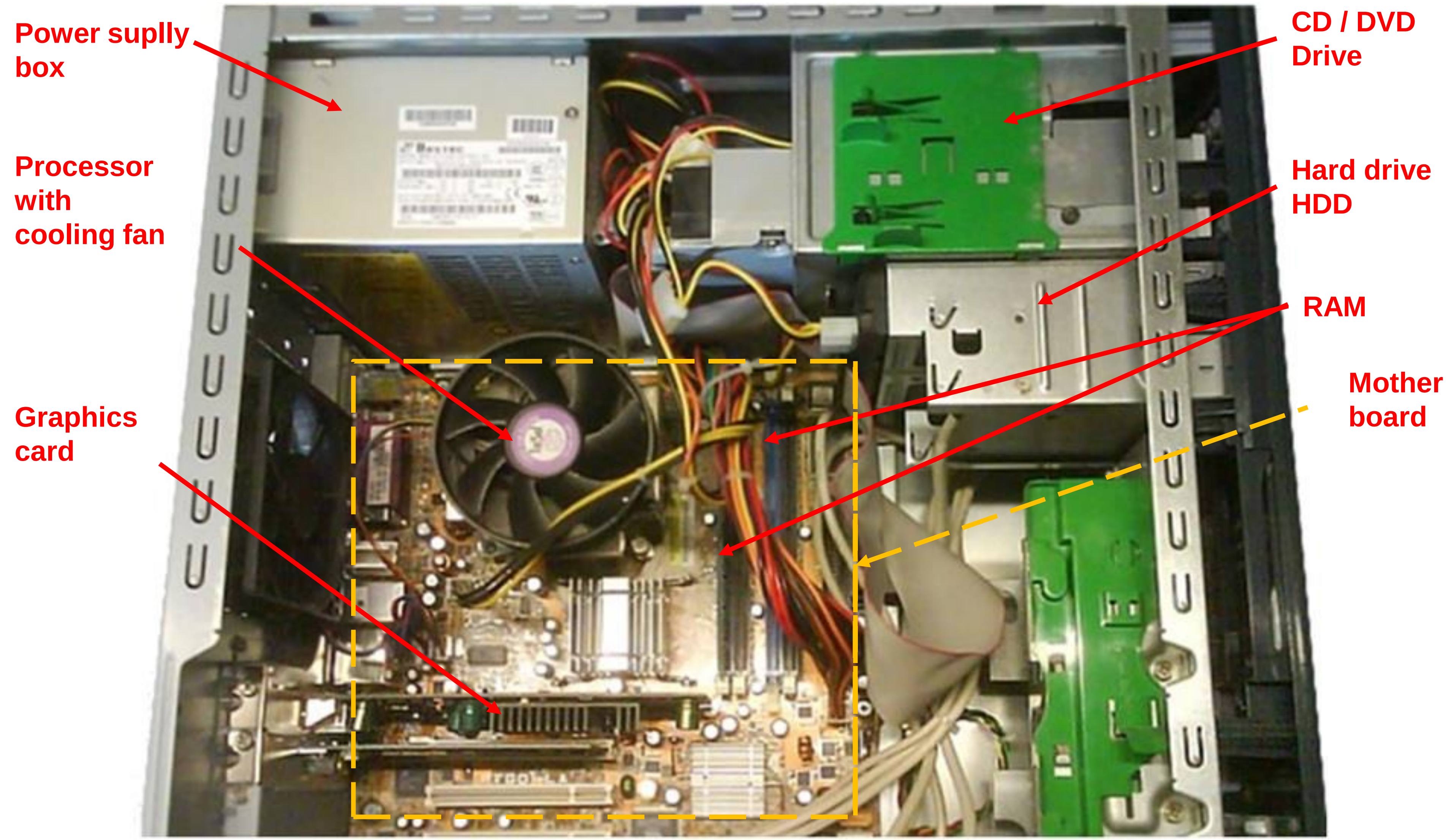
Hardware: Main Components of a PC

- **Central unit (the case)**
  - Processor or CPU (Central Processing Unit)
  - Main memory (RAM)
  - Storage media (HDD, SSD, CD/DVD, flash drive, etc.)
  - Specialized cards (video cards, network cards, etc.)
  - Input/Output Interfaces (Ports)
- **Peripherals**
  - Input: keyboard, mouse, microphone,…
  - Output: screen, printer, speakers,…
  - Input/Output: modem, headphones,…

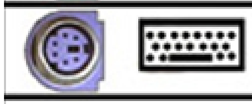
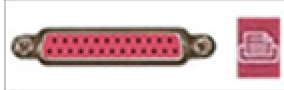
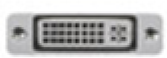
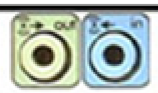
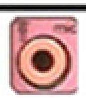
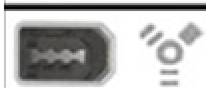
Hardware: The Central Unit



Hardware: The Central Processing Unit



Hardware: Ports (peripheral connections)

Connectors	Descriptions
	Mouse
	Keyboard
	USB port for devices with USB connectors
	Printer
	Monitor
	Audio input and output lines
	Microphone
	Headphones
	FireWire (IEEE 1394) for cameras or other very high-speed peripherals

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What Is An Operating System?

- **A Operating System is a set of programs that manage hardware and control applications**
  - Device management (screen display, keyboard reading, printer control, etc.)
  - User and data management (accounts, resource sharing, file and directory management, etc.)
  - User interface (textual or graphical): Interpretation of commands
  - Program control (task splitting, CPU time sharing, etc.)

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Computer languages

- **A computer language is a tool used to give orders (instructions) to the machine.**
  - Each instruction corresponds to a processor action
- **Purpose: write programs (a consecutive sequence of instructions) designed to perform a given task**
  - Example: a pharmaceutical product management program
- **Constraint: must be understandable by the machine**

High Level Computer Languages

- **Multiple benefits for the High Level:**
  - Close to human language "English" (understandable)
  - Allows for greater portability (independent from hardware)
  - Manipulation of complex data and expressions (reals, objects, a*b/c, ...)

- **Need for a translator (Compiler/Interpreter)**

Source code in advanced language → Compiler or interpreter → **Binary language**

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User Applications

- **Office applications:**
  - Word (Writing)
  - Excel (Calculation)
  - PowerPoint (Presentation)

- **Communication and internet applications :**
  - Messaging
  - Social networks
  - Web browsers

Medical Applications (1)

- **Medical database management**
  - Patients, Medications, Diseases
- **Medical facility management**
  - Hospitals, Pharmacies, Labs, Doctors' offices
- **Control, monitoring and prevention**
  - Epidemic Surveillance Center
  - Pharmacovigilance

Medical Applications (2)

- **Artificial intelligence**
  - Medical diagnosis
- **Virtual medical simulation**
  - Training and learning in surgery
- **Medical imaging**
  - X-rays, Ultrasounds, MRI, and CT Scans
- **Telemedicine**
  - Remote monitoring, teleconsultation, telesurgery

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  - Data size measurement units

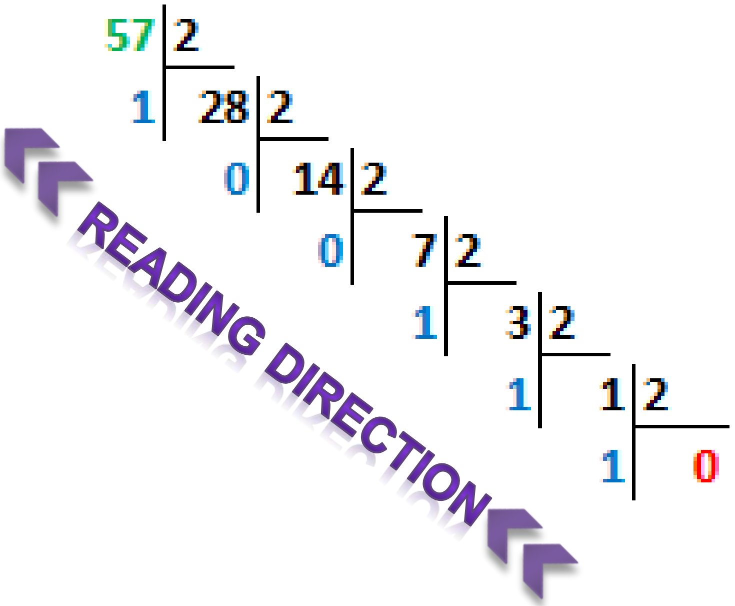
Data representation (1)

- **On computers, data is represented and saved on physical media (memory) in binary.**
- **This is not a random choice, but a current technological limitation.**
- **Constraint: everything must be represented using the symbols 0 and 1**
  - Text
  - Picture
  - Sound
  - ... Etc.

Data representation (2)

- **Converting decimal to binary**
  - Perform successive integer divisions by 2 until the quotient becomes zero.
- **Conversion example:**
  - 57 (decimal base) = 111001 (binary base)

$57_{(10)} = 111001_{(2)}$



Data representation (3)

- **Converting binary to decimal**
  - To convert a binary number to a decimal number, you must multiply the value of each bit by its weight, then add the results together.
- **Conversion example:**
  - 11010 (2) = 1x2 ^{4} + 1x2 ^{3} + 0x2 ^{2} + 1x2 ^{1} + 0x2 ^{0} = 26 (10)

Rank	4	3	2	1	0
weight	2^4	2^3	2^2	2^1	2^0
	1	1	0	1	0

= $1x2^4 + 1x2^3 + 0x2^2 + 1x2^1 + 0x2^0$

= 16 + 8 + 0 + 2 + 0

= 26

Data size measurement units

- The primary unit of measurement for memory size is the byte (8 bits).
- A bit is the elementary binary unit (0 or 1)

Unit name	Designation	Value and equivalent
Bit	b	2^0 b = 1 bit (Elementary unit)
Byte	B	2^3 b = 8 bits = 1 Byte (Primary Unit)
Kilobyte	KB	2^{10} = 1024 bytes
Megabyte	MB	2^{20} B = 2^{10} KB
Gigabyte	GB	2^{30} B = 2^{10} MB = 2^{20} KB
Terabyte	TB	2^{40} B = 2^{10} GB = 2^{20} MB = 2^{30} KB
Petabyte	PB	2^{50} B = 2^{10} TB = 2^{20} GB = 2^{30} MB = 2^{40} KB